

ECON601 TA Notes: Primal Approach to Optimal Policy

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1 Optimal Policy: Static Ramsey Problem

A **Ramsey planner** is benevolent. The planner is subject to technology and expenditure shocks, and the goal is to finance government expenditure using only distortionary taxes, with no lump-sum taxes. This is a general-equilibrium setting with implementability requirements; that is, agents in the economy must have an incentive to realize the allocation.

1.1 (Revisiting) Definitions and Environments

First, we review the notation:

- $\mathbf{p}$ is the price vector
- $\mathbf{z}$ is the exogenous variable vector (technology shocks, government expenditures)
- $\boldsymbol{\pi}$ is the government policies (taxes)
- $\mathbf{x}$ is the allocation

Definition 1. A **competitive equilibrium** is $(\mathbf{p}, \mathbf{x})$ given $(\mathbf{z}, \boldsymbol{\pi})$, where $\mathbf{x}$ is feasible, optimal for the agents in the economy given $(\mathbf{p}, \mathbf{z}, \boldsymbol{\pi})$ and markets clear.

Definition 2. The solution to the **social planner's problem** is an allocation $\mathbf{x}$ which is feasible and maximizes the social welfare function given $\mathbf{z}$

Definition 3. The solution to the **Ramsey planner's problem** is $(\mathbf{p}, \boldsymbol{\pi}, \mathbf{x})$ where $\mathbf{x}$ maximizes the social welfare function given $\mathbf{z}$, subject to $(\mathbf{p}, \mathbf{x})$ being a competitive equilibrium given $(\mathbf{z}, \boldsymbol{\pi})$

Proposition 1. *If $\mathbf{x}$ is a solution to the social planner's problem given $\mathbf{z}$ and there is a $\mathbf{p}, \boldsymbol{\pi}$ where $(\mathbf{p}, \mathbf{x})$ is a competitive equilibrium given $\mathbf{z}$ and $\boldsymbol{\pi}$, then $(\mathbf{p}, \boldsymbol{\pi}, \mathbf{x})$ is a solution to the Ramsey planner's problem given $\mathbf{z}$*

Now, for the environment of the economy, we assume that there are n goods in the economy produced by labor using a production function

$$F(x_1, \dots, x_n, h) \leq 0$$

- x_i : output of good i ;
- h : labor;

- F : a constant-returns-to-scale (CRS) technology;
- Possibility of intermediate goods
- One-good example: $F(x, h) = x - h \leq 0$.

Consumer's problem:

$$\max_{c_1, \dots, c_n, h} u(c_1, \dots, c_n, 1 - h) \quad (1)$$

subject to

$$\sum_i p_i(1 + \tau_i)c_i = h \quad (2)$$

Note that here we assume:

- τ_i is the consumption tax on consumption good i ;
- p_i is the price of good i ;
- the wage rate is normalized to one;
- the CRS technology implies zero profits for the firm, which are therefore omitted below.

Firm's Problem:

$$\max_{\mathbf{x}, h} \sum_i p_i x_i - h \quad (3)$$

subject to

$$F(\mathbf{x}, h) \leq 0 \quad (4)$$

Because of the CRS assumption, firms earn zero profits.

$$\sum_i p_i x_i = h \quad (5)$$

Government's budget constraint is:

$$\sum_i p_i g_i = \sum_i p_i \tau_i c_i \quad (6)$$

where $(g_i)_{i=1}^n$ are given.

Market clearing requires that for $i = 1, 2, \dots, n$:

$$c_i + g_i = x_i \quad (7)$$

1.2 Competitive Equilibrium

Now, we can formally define competitive equilibrium in this environment.

Definition 4. A **competitive equilibrium** is a policy $\boldsymbol{\pi} = (\tau_i)_{i=1}^n$, a set of exogenous government expenditures $\mathbf{g}$, allocations $\mathbf{c}, h, \mathbf{x}$, and a price system $\mathbf{p}$ such that the following conditions hold:

- Given $\boldsymbol{\pi}$ and $\mathbf{p}$, $\mathbf{c}$ and h solve the consumer's problem (1), subject to (2).
- Given $\mathbf{p}$, $\mathbf{x}$ and $\mathbf{h}$ solve the firm's problem (3) subject to (4)
- $\boldsymbol{\pi}, \mathbf{p}, \mathbf{g}, \mathbf{c}$ satisfy the government budget constraint (6)
- $\mathbf{c}, \mathbf{g}$, and $\mathbf{x}$ satisfy the market-clearing condition (7) for every good i .

We assume all necessary concavity, monotonicity, and Inada conditions to ensure that the FOCs are necessary and sufficient.

Household Lagrangian is:

$$\mathcal{L}_h = u(c_1, \dots, c_n, 1 - h) - (\lambda \sum_i p_i(1 + \tau_i)c_i - h)$$

The first-order condition yields:

$$u_i = \lambda(1 + \tau_i)p_i, \forall i = 1, \dots, n \quad (8)$$

$$u_l = \lambda \quad (9)$$

$$\sum_i p_i(1 + \tau_i)c_i = h \quad (10)$$

This leads to the characterization of:

$$\frac{u_i}{u_j} = \frac{p_i}{p_j} \frac{1 + \tau_i}{1 + \tau_j} \quad (11)$$

$$\frac{u_i}{u_l} = p_i(1 + \tau_i) \quad (12)$$

$$\sum_i p_i(1 + \tau_i)c_i = h \quad (13)$$

For the firm,

$$\mathcal{L}_f = \sum_i p_i x_i - h - \mu F(x_1, x_2, \dots, x_n, h)$$

FOC yields:

$$p_i = \mu F_i \quad (14)$$

$$-1 = \mu F_h \quad (15)$$

Thus, we have

$$p_i = -\frac{F_i}{F_h} \forall i = 1, 2, \dots, n \quad (16)$$

$$F(x_1, x_2, \dots, x_n, h) = 0 \quad (17)$$

Then the full characterization of a competitive equilibrium is as follows:

Definition 5. (Competitive Equilibrium) Given $\{\tau_i, g_i\}$, the equilibrium is a sequence $\{x_i, c_i, p_i\}_{i=1}^n$ and h such that

$$\frac{u_i}{u_l} = p_i(1 + \tau_i) \forall i = 1, 2, \dots, n \quad (18)$$

$$\sum_i p_i(1 + \tau_i)c_i = h \quad (19)$$

$$p_i = -\frac{F_i}{F_h} \forall i = 1, 2, \dots, n \quad (20)$$

$$F(x_1, x_2, \dots, x_n, h) = 0 \quad (21)$$

$$c_i + g_i = x_i \forall i = 1, 2, \dots, n \quad (22)$$

1.3 Optimal Policy

Let Π be the set of policies that the government can choose π from for which a CE exists. For every $\pi \in \Pi$, the allocations $\mathbf{c}(\pi')$, $\mathbf{x}(\pi')$, $h(\pi')$ and price $\mathbf{p}(\pi')$ and the policy π' constitute a competitive equilibrium, given $\mathbf{g}$. Now a formal definition of Ramsey equilibrium.

Definition 6. A **Ramsey Equilibrium** is a policy $\pi \in \Pi$, allocation rules $\mathbf{c}(\pi')$, $\mathbf{x}(\pi')$, $h(\pi')$ and price $\mathbf{p}(\pi')$ such that policy π solves $\max_{\pi' \in \Pi} u(\mathbf{c}(\pi'), h(\pi'))$.

Alternatively, given $\mathbf{g}$,

$$\{(\tau_i)_{i=1}^n, (c_i)_{i=1}^n, (x_i)_{i=1}^n, (p_i)_{i=1}^n, h\} = \arg \max_{\pi, \mathbf{p}, \mathbf{x}, \mathbf{c}, h} u(c_1, c_2, \dots, c_n, 1 - h) \quad (23)$$

subject to (18) - (22).

Ramsey Planner's Problem: An Overview Generally, there are two approaches to solving (23):

- Dual approach: choose policies under which households optimize and social welfare is maximized.
- Primal approach: convert the problem into one in which the only choice objects are allocations and equilibrium is represented by the **implementability constraint**.

We are going to illustrate how to solve the Ramsey problem by the primal approach. The basic idea is to express the price and policy vectors in terms of allocations and do the Ramsey planner's maximization problem (23) in the allocation space only. In order to implement this idea, we need to understand **what conditions make a particular allocation $\mathbf{x}$ part of a competitive equilibrium**. It sounds complicated, but eventually we will figure out that $\mathbf{x}$ only needs to satisfy two conditions: **feasibility** and **implementability**, to make itself part of a competitive equilibrium.

Derivation of Feasibility Constraint We define the **feasibility constraint** by substituting (22) into (21):

$$F(c_1 + g_1, \dots, c_n + g_n, h) = 0 \quad (24)$$

Note:

- g_i are not considered as policy in the current setup; they are taken as exogenous variables;
- Essentially, this is the resource constraint.

Derivation of Implementability Constraint We define the **implementability constraint** by substituting (18) into (19) and multiplying both sides by u_l :

$$\sum_i u_i c_i - u_l h = 0 \quad (25)$$

Note:

- Here, we replace the price-and-policy terms $p_i(1 + \tau_i)$ in the consumer's budget constraint with the marginal rate of substitution between good i and leisure, which depends only on allocations.
- This is necessary for the allocation to be part of a competitive equilibrium because equilibrium prices must equal the corresponding marginal rates of substitution.

Where have all the prices gone? Before discussing the Ramsey planner's optimization problem, I explain how to recover prices from allocations. For an allocation to be part of a competitive equilibrium, there must be a price vector that justifies it.

It turns out that it is sufficient for us to know the optimal allocation only because as long as we know what the optimal allocation is, we can back out prices and optimal policies easily from the optimal allocations:

- First, notice that Equation (20), the optimality condition for the firm (reads “marginal rate of technological substitution equals to relative price of inputs”), allows us to back out the price vector $\mathbf{p}$ from the allocation $\mathbf{x}$.
- Furthermore, combining (18) and (20), we can back out the optimal policy from optimal allocation:

$$1 + \tau_i = -\frac{F_h}{F_i} \frac{u_i}{u_l}, \forall i = 1, 2, \dots, n \quad (26)$$

We usually call (26) the policy that implements optimal allocation in a competitive equilibrium, or simply “optimal policy”.

Alternatively, the **implementability constraint** can also be derived from government spending side. Note that we can combine $c_i + g_i = x_i$ with $\sum_i p_i \tau_i c_i = \sum_i p_i g_i$, and get

$$\sum_i p_i x_i = \sum_i p_i (1 + \tau_i) c_i$$

Combine this with (4.18) and $\sum_i p_i x_i = h$ (4.5), the CRS result from production function F , we get:

$$\sum_i \frac{u_i}{u_l} c_i = h$$

Multiply both sides by u_l we get:

$$\sum_i u_i c_i - u_l h = 0$$

Now, we can leave out $\boldsymbol{\tau}$ and $\mathbf{p}$ as choice variables from the problem because now given other variables, those two are defined through the constraints. The following results are immediate:

Proposition 2. *(Necessity and sufficiency of feasibility and implementability.) Any allocation $\mathbf{x}$ in the CE satisfies the feasibility constraint (24) and the implementability constraint (25). Furthermore, given allocations that satisfy (24) and (25), we can find a price system $\mathbf{p}$*

and a policy π such that allocation $\mathbf{x}$ is part of a competitive equilibrium under price $\mathbf{p}$ and policy π . Prices are defined by (20), and policy π is defined by (26).

Proposition 3. *The Ramsey allocations $(\mathbf{c}, h)$ solve the Ramsey problem, which is to choose $\mathbf{c}, h$ to maximize $u(\mathbf{c}, 1 - h)$ subject to the feasibility constraint (24) and the implementability constraint (25).*

1.4 Solving Ramsey Problem

Ramsey problem is:

$$\max_{c_1, \dots, c_n, h} u(c_1, \dots, c_n, h)$$

subject to

$$\begin{aligned} F(c_1 + g_1, \dots, c_n + g_n, h) &= 0 \\ \sum_i u_i c_i - u_l h &= 0 \end{aligned}$$

We assign λ to IC, and γ to FC. Lagrangian:

$$\mathcal{L}_{Ramsey} = u(c_1, \dots, c_n, h) - \lambda \left(\sum_i u_i c_i - u_l h \right) - \gamma F(c_1 + g_1, \dots, c_n + g_n, h)$$

FOC yields:

$$\begin{aligned} u_i &= \gamma F_i + \lambda (u_{1i} c_1 + \dots + (u_{ii} c_i + u_i) + \dots + u_{ni} c_n - u_{li} h) \\ &= \gamma F_i + \lambda \sum_j u_{ji} c_j + \lambda u_i - \lambda u_{li} h \\ u_l &= \lambda \sum_j u_{jl} c_j + \lambda (u_l - u_{ll} h) - \gamma F_h \end{aligned}$$

These can be simplified to

$$(1 - \lambda) u_i - \lambda \sum_j u_{ji} c_j + \lambda u_{li} h - \gamma F_i = 0 \quad (27)$$

$$-(1 - \lambda) u_l + \lambda \sum_j u_{jl} c_j - \lambda u_{ll} h - \gamma F_h = 0 \quad (28)$$

For algebraic simplicity, we define:

$$H_i = -\frac{\sum_j u_{ji}c_j - u_{li}h}{u_i} \quad (29)$$

$$H_l = \frac{\sum_j u_{jl}c_j - u_{ll}h}{u_l} \quad (30)$$

This simplifies (27), (28) to

$$(1 - \lambda)u_i + \lambda H_i u_i = \gamma F_i \quad (31)$$

$$-(1 - \lambda)u_l + \lambda H_l u_l = \gamma F_h \quad (32)$$

There are 4 results that we will go into some details:

- If $H_i > H_j$, then $\tau_i > \tau_j$
- If the utility function is separable and linear in leisure (quasi-linear), goods whose price elasticities of demand are higher are taxed less, i.e. GE result same as PE result
- **Uniform commodity taxation** If c_i enter the utility function homothetically, then $\tau_i = \tau$ for all i
- **Non-Taxation of Intermediate Goods:** It is optimal to not tax intermediate goods

Result 1

By combining (31) and (32), we get:

$$\begin{aligned} \frac{(1 - \lambda + \lambda H_i)u_i}{(1 - \lambda - \lambda H_l)u_l} &= -\frac{F_i}{F_h}, \forall i = 1, \dots, n \\ \implies \left(1 + \frac{\lambda(H_i + H_l)}{1 - \lambda - \lambda H_l}\right) \frac{u_i}{u_l} &= -\frac{F_i}{F_h} \end{aligned}$$

Recall from (26), we have

$$1 + \tau_i = -\frac{F_h u_i}{F_i u_l}, \forall i = 1, 2, \dots, n$$

Combining these we get the condition under which optimal policy τ_i must satisfy:

$$\begin{aligned} 1 + \frac{\lambda(H_i + H_l)}{1 - \lambda - \lambda H_l} &= \frac{1}{1 + \tau_i} \\ \implies \frac{\lambda(H_i + H_l)}{1 - \lambda - \lambda H_l} &= -\frac{\tau_i}{1 + \tau_i} \end{aligned}$$

Then take $j \neq i$, we have:

$$\begin{cases} \frac{\lambda(H_i + H_l)}{1 - \lambda - \lambda H_l} = -\frac{\tau_i}{1 + \tau_i} = \frac{1}{1 + 1/\tau_i} \\ \frac{\lambda(H_j + H_l)}{1 - \lambda - \lambda H_l} = -\frac{\tau_j}{1 + \tau_j} = \frac{1}{1 + 1/\tau_j} \end{cases}$$

Hence

$$\frac{H_i + H_l}{H_j + H_l} = \frac{1 + 1/\tau_j}{1 + 1/\tau_i}$$

Hence, it is clear that if $H_i > H_j$, we have $\tau_i > \tau_j$.

Result 2: Higher Taxes on Goods with Inelastic Demand

By assumption, the utility function takes the form of:

$$u(c_1, \dots, c_n, 1 - h) = \sum_i v^i(c_i) + (1 - h)$$

Then by our definition,

$$H_i = -\frac{v_i''(c_i)c_i}{v_i'(c_i)}$$

since $u_{ji} = 0$ for $i \neq j$, by additive separability.

Recall that the price elasticity of demand for good i is defined as:

$$\epsilon_p^i = -\frac{\frac{\partial c_i}{c_i}}{\frac{\partial p_i}{p_i}}$$

Now, going back to CE result, we have the household's FOC being:

$$v_i'(c_i) = \lambda(1 + \tau_i)p_i \tag{33}$$

$$1 = \lambda \tag{34}$$

From $\lambda = 1$ (from your micro class: if you work with a quasi-linear utility, then all the income effects are absorbed by the linear good, which is essentially what $\lambda = 1$ is saying),

we plug it into the FOC on c_i and apply implicit function theorem with respect to p_i :

$$\begin{aligned}
& \frac{\partial v'_i(c_i)}{\partial c_i} \cdot \frac{\partial c_i}{\partial p_i} = (1 + \tau_i) \\
\implies & v''_i(c_i) \cdot \frac{\partial c_i}{\partial p_i} = (1 + \tau_i) \\
\implies & v''_i(c_i) \cdot \underbrace{\frac{\partial c_i}{\partial p_i} \cdot \frac{p_i}{c_i}}_{\equiv -\epsilon_p^i} = \underbrace{(1 + \tau_i) \frac{p_i}{c_i}}_{=v'(c_i)/c_i} \\
\implies & \epsilon_p^i = -\frac{1}{\frac{v''_i(c_i)c_i}{v'_i(c_i)}} = \frac{1}{H_i}
\end{aligned}$$

Then, by Result 1, if good i is more price elastic than good j , then $\epsilon_p^i > \epsilon_p^j > 0$, then $0 < H_i < H_j$. Then $\tau_i < \tau_j$. The government should levy higher taxes on more demand-inelastic goods.

Result 3: Uniform Taxation under Homothetic Utility

Suppose the utility function is in the following form:

$$u(c_1, \dots, c_n, 1 - h) = W(G(c_1, \dots, c_n), 1 - h)$$

where G satisfies homotheticity.

With this utility function we have:

$$\frac{u_i(\alpha \mathbf{c}, 1 - h)}{u_k(\alpha \mathbf{c}, 1 - h)} = \frac{u_i(\mathbf{c}, 1 - h)}{u_k(\mathbf{c}, 1 - h)}$$

for any $\alpha > 0$.

Recall that from the household's problem and firm's problem,

$$\begin{aligned}
\frac{u_i}{u_l} &= p_i(1 + \tau_i) \\
p_i &= -\frac{F_i}{F_h}
\end{aligned}$$

This gives

$$1 + \tau_i = -\frac{u_i}{u_l} \frac{F_h}{F_i}$$

We can also do this trick:

$$u_i(\alpha \mathbf{c}, 1 - h) = \frac{u_i(\mathbf{c}, 1 - h)}{u_k(\mathbf{c}, 1 - h)} u_k(\alpha \mathbf{c}, 1 - h)$$

Also note that we have:

$$\sum_{j=1}^n u_{ij} c_j = \frac{u_i(\mathbf{c}, 1 - h)}{u_k(\mathbf{c}, 1 - h)} \sum_{j=1}^n u_{kj} c_j$$

Then, $\forall i, k$,

$$\frac{\sum_{j=1}^n u_{ij} c_j}{u_i} = \frac{\sum_{j=1}^n u_{kj} c_j}{u_k} := A$$

where A is just a constant.

Now, we have, from Ramsey's problem:

$$(1 - \lambda)u_i - \lambda \left(\sum_{j=1}^n u_{ij} c_j - u_{li} h \right) = \gamma F_i$$

Recall our setup on the utility function, we have the following derivatives:

$$u_i = W_1 G_i$$

$$u_{li} = W_{12} G_i$$

Substituting those back in and we get:

$$(1 - \lambda)W_1 G_i - \lambda(AW_1 G_i - W_{12} G_i h) = \gamma F_i$$

$$\frac{G_i}{F_i} = \frac{\gamma}{(1 - \lambda)W_1 - \lambda AW_1 + \lambda W_{12} h}$$

Note that this gives

$$\frac{G_i}{G_j} = \frac{F_i}{F_j} = \frac{u_i}{u_j}$$

Thus, we have

$$\tau_i = \tau_j = \tau, \forall i, j$$

Result 4: No Taxation for Intermediate Goods

Consider the setup where there are 2 firms

- intermediate good producer with production function $k(z, h_2) \leq 0$, with z being intermediate good output and h_2 being the labor input

- final good producer with production function $f(x, z, h_1) \leq 0$, with x being the final good output, z being the input and h_1 being the labor input

Consumer takes the same form $u(c, 1 - h)$, and market clearing is $c + g = x$ and $h = h_1 + h_2$. To derive the implementability constraint, we see that the household yields the same result:

$$u_c c - u_l(h_1 + h_2) \leq 0$$

For the final good producer, we have

$$\max_{x, z, h_1} px - wh_1 - q(1 + \eta)z$$

subject to $f(x, z, h_1) \leq 0$.

The condition yields:

$$\frac{f_z}{f_h} = \frac{q(1 + \eta)}{w} \quad (35)$$

For the intermediate good producer, we have

$$\max_{z, h_2} qz - wh_2$$

subject to $k(z, h_2) \leq 0$.

The FOC yields:

$$\frac{k_z}{k_h} = -\frac{q}{w} \quad (36)$$

With (35) and (36) we get:

$$\frac{f_z}{f_h} = (1 + \eta) \frac{k_z}{k_h} \quad (37)$$

Now going back to the Ramsey's problem, writing out the Lagrangian and we get:

$$\mathcal{L} = u(x, 1 - h_1 - h_2) - \lambda(u_x x - u_l(h_1 + h_2)) - \theta f(x, z, h_1) - \mu k(z, h_2)$$

FOC for z, h_1, h_2 yields:

$$\begin{aligned} -\theta f_z - \mu k_z &= 0 \\ -u_l - \lambda(-u_{xl}x + u_{ll}(h_1 + h_2)) &= \theta f_h \\ -u_l - \lambda(-u_{xl}x + u_{ll}(h_1 + h_2)) &= \mu k_h \end{aligned}$$

Combining those together, we get $\frac{f_z}{f_h} = \frac{k_z}{k_h}$. Hence $\eta = 0$. The intermediate good is not taxed.

1.5 Social Planner's Problem

Now, for social planner's problem, we have:

$$\max_{c_1, \dots, c_n, h} u(c_1, \dots, c_n, 1 - h) - \gamma F(c_1 + g_1, \dots, c_n + g_n, h) \quad (38)$$

FOC yields:

$$\begin{aligned} u_i &= \gamma F_i \\ -u_l &= \gamma F_h \end{aligned}$$

This gives:

$$\frac{u_i}{u_l} = -\frac{F_i}{F_h} \quad (39)$$

Recall that Ramsey FOC before

$$\begin{aligned} (1 - \lambda)u_i + \lambda H_i u_i &= \gamma F_i \\ -(1 - \lambda)u_l + \lambda H_l u_l &= \gamma F_h \end{aligned}$$

Define

$$1 - \omega_i = \frac{1 - \lambda - \lambda H_l}{1 - \lambda + \lambda H_i} \quad (40)$$

We have the Ramsey FOC yields:

$$\frac{u_i}{u_l} = (1 - \omega_i) \frac{-F_i}{F_h} \quad (41)$$

Comparing social planner solution and Ramsey planner solution, we see that Ramsey planner wants to create a **wedge** between all *MRS* and *MRT* pairs, and wedge is defined by (26):

$$1 + \tau_i = -\frac{F_h}{F_i} \frac{u_i}{u_l}, \forall i = 1, 2, \dots, n$$

Hence

$$1 - \omega_i = 1 + \tau_i \quad (42)$$

1.6 Completeness of the Tax System

Suppose there is a labor income tax. The equilibrium condition is:

$$\frac{u_i}{u_l} = \frac{p_i(1 + \tau_i)}{1 - \tau_h}$$

This yields that for $i = 1, 2, \dots, n$,

$$\frac{1 + \tau_i}{1 - \tau_h} = -\frac{u_i}{u_l} \frac{F_h}{F_i}$$

Then there is a problem to pin down all $n + 1$ tax rate since we only have n equations:

$$1 - \omega_i = \frac{1 + \tau_i}{1 - \tau_h}, \forall i = 1, \dots, n$$

Now, suppose we set one of the instrument exogenously, for example, $\tau_1 = 0$. Then the equilibrium involves

$$-\frac{F_1}{F_h} = \frac{u_1}{u_l}$$

Now the system has enough equations to solve the n unknowns. We need to add this as a constraint on the Ramsey planner's problem.

Remark: The Ramsey problem is **not** about taxation per se, but about creating wedges between MRS–MRT pairs. The Ramsey solution specifies the optimal wedges or distortions given the government's financing (spending) requirement. There are n margins in the economy, and the Ramsey planner wants to create wedges in all of them.

We will end the discussion of static Ramsey problem with a proposition describing when Ramsey problem's solution coincides with social planner's solution.

Proposition 4. *If $g_i > 0$ for at least some $i = 1, \dots, n$, then the IC multiplier $\lambda \neq 0$. If $g_i = 0$ for all $i = 1, \dots, n$, then $\lambda = 0$, then the solution to the Ramsey problem coincides with the solution to the social planner's problem.*

2 Ramsey Application: Optimal Fiscal Policy

2.1 Setup

We use the standard one-sector neoclassical growth model in a real environment (without money).

There are 2 sources of uncertainty:

- technology shocks
- government spending

Here are the notations for the uncertainty setup:

- each period, the economy experiences one of the finitely many events $s_t \in S = \{e_1, \dots, e_n\}$
- history up to time t is defined as $s^t = (s_0, s_1, \dots, s_t) \in S \times S \times \dots \times S$ ($t+1$ times) and the probability as of $t=0$, of a particular history s^t is $\mu(s^t)$
- initial realization s_0 taken as given
- we simplify the notation from Chari and Kehoe (1999) by using $x_t := x(s^t)$.

Consumer Preferences of consumers are

$$\max_{c_t, h_t, k_t, b_t} \mathbb{E}_0 \sum_t \beta^t u(c_t, 1 - h_t) \quad (43)$$

The budget constraint is:

$$c_t + k_t + b_t \leq (1 - \tau_t^h)w_t h_t + R_t^k k_{t-1} + R_t^b b_{t-1} \quad (44)$$

where

- $R_t^k = 1 + (1 - \tau_t^k)(r_t - \delta)$ is the return to capital
- τ_t^k and τ_t^h are the tax rates on incomes from capital and labor
- b_t is the stock of government debt the consumer holds at time t (and state s^t), with return R_{t+1}^b

Firm Firm's production at time t is given by $z_t F(k_{t-1}, h_t)$, where F is a CRS production function and z_t is the technology shock.

Firm pays their marginal costs to factors of production

$$r_t = z_t F_k(k_{t-1}, h_t) \quad (45)$$

$$w_t = z_t F_h(k_{t-1}, h_t) \quad (46)$$

Government The government's budget constraint is given by

$$b_t = R_t^b b_{t-1} + g_t - \tau_t^h w_t h_t - \tau_t^k (r_t - \delta) k_{t-1} \quad (47)$$

where g_t is the stochastic government spending shock. This is essentially the market clearing condition for bond.

Market Clearing Since we already (implicitly) required the bond, labor and capital market clearing in the government budget constraint, we no longer need to specify goods market clearing condition to define a competitive equilibrium. However, for the purpose of Ramsey Planner's problem and Social Planner's Problem, we still derive the goods market clearing condition/feasibility constraint/resource constraint:

$$c_t + g_t + k_t = z_t F(k_{t-1}, h_t) + (1 - \delta) k_{t-1} \quad (48)$$

Lastly, the laws of motion for the stochastic variables are:

$$[z_t, g_t] = \chi(z_{t-1}, g_{t-1}; \epsilon_t)$$

where ϵ_t are i.i.d. shocks.

Competitive Equilibrium

A few additional notations:

- $x_t = (c_t, h_t, k_t, b_t)$ denotes an allocation for consumer at t (and s^t), and $x = \{x_t\}_{t=0}^\infty$ denotes an allocation for all t and s^t chosen at time 0
- $p_t = (w_t, r_t, R_t^b)$ denotes a price system at time t and $p = \{p_t\}_{t=0}^\infty$ denotes a price for all t and s^t chosen at time 0

- $\pi_t = (\tau_t^h, \tau_t^k)$ denotes the government policy at t and $\pi = \{\pi_t\}_{t=0}^\infty$ denotes an infinite sequence of policies for all t and s^t chosen at time 0
- initial stock of debt and capital b_{-1} and k_{-1} are given.

Definition 7. A competitive equilibrium for this economy is an allocation x , and a price system p , given a policy π such that:

- Given π and p , x maximizes (43) subject to the sequence of budget constraint (44)
- p satisfies (45) and (46)
- government budget constraint (47) is satisfied.

The above definition already stipulates labor, capital and bond market clearing, which implies goods market clearing condition (48).

Write them in equations: given $\{z_t, g_t, \tau_t^h, \tau_t^k\}$, allocation $\{c_t, h_t, b_t, k_t\}$ prices $\{w_t, r_t, R_t^b\}$ (in total 7 variables):

$$u_l(t) = u_c(t)(1 - \tau_t^h)w_t \quad (49)$$

$$u_c(t) = \beta \mathbb{E}_t[u_c(t+1)R_{t+1}^k] \quad (50)$$

$$u_c(t) = \beta \mathbb{E}_t[u_c(t+1)R_{t+1}^b] \quad (51)$$

$$r_t = z_t F_k(k_{t-1}, h_t) \quad (52)$$

$$w_t = z_t F_h(k_{t-1}, h_t) \quad (53)$$

$$c_t + g_t + k_t = z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} \quad (54)$$

$$b_t = R_t^b b_{t-1} + g_t - \tau_t^h w_t h_t - \tau_t^k (r_t - \delta)k_{t-1} \quad (55)$$

and the Transversality conditions:

$$\lim_{t \rightarrow \infty} \beta^t u_c(t) k_t = 0$$

$$\lim_{t \rightarrow \infty} \beta^t u_c(t) b_t = 0$$

2.2 Ramsey Problem: Main Result

Before solving the model, we first recognize two potential problems raised by introducing dynamics:

- **time inconsistency:** in any period t , capital was predetermined in $t - 1$, given s^{t-1} and all government policies π rather than only $\pi(s^t)$. From the perspective of period t , the government appears to have an incentive to tax capital at 100% because doing so will not distort a predetermined decision. Thus, the period- t government will find the policy set at $t = 0$ suboptimal. To sidestep this issue, we assume that the government can commit perfectly and credibly to its policy at $t = 0$.
- **tax in period 0:** because k_{-1} is in fixed supply, it is optimal for the government to tax it as much as possible. To avoid this issue, we assume that τ_0^k is fixed at a specified rate.

Definition 8. A Ramsey equilibrium is a policy π , an allocation rule $x()$ and prices $p()$ such that

$$\pi = \arg \max_{\pi'} \mathbb{E}_0 \sum_t \beta^t u(c_t(\pi'), 1 - h_t(\pi'))$$

where for every π' , the allocation $x(\pi')$ and prices $p(\pi')$ constitute a competitive equilibrium given the policy π' .

Again, to use the primal approach to find the optimal policy, we must characterize the conditions under which an allocation can be supported as part of a competitive equilibrium. The following proposition is the main result and is essentially the dynamic counterpart of the static problem considered above:

Proposition 5. *The allocation x , capital tax rate and return on debt in period 0 in a competitive equilibrium satisfy the resource constraint*

$$c_t + g_t + k_t = z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} \quad (56)$$

and the **Present Value Implementability Constraint (PVIC)**

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [u_c(c_t, 1 - h_t)c_t - u_l(c_t, 1 - h_t)h_t] = u_c(c_0, 1 - h_0)[R_0^k k_{-1} + R_0^b b_{-1}] \quad (57)$$

Furthermore, given allocations and period-0 policies that satisfy (56) and (57), we can construct policies, prices, and debt holdings that, together with the given allocations and period-0 policies, constitute a competitive equilibrium.

2.2.1 Derivation of PVIC

(Lixi: This is one of the most important subjects covered in this course.)

Starting from the consumer's budget constraint in each period and multiplying by the Lagrange multiplier $\beta^t u_c(c_t, 1 - h_t)$, or simply $\beta^t u_c(t)$, we obtain

$$\mathbb{E}_0 \left\{ \underbrace{\sum_{t=0}^{\infty} \beta^t u_c(t) (1 - \tau_t^h) w_t h_t}_{\text{I}} + \underbrace{\sum_{t=0}^{\infty} \beta^t u_c(t) R_t^k k_{t-1}}_{\text{II}} + \underbrace{\sum_{t=0}^{\infty} \beta^t u_c(t) R_t^b b_{t-1}}_{\text{III}} \right. \\ \left. - \underbrace{\sum_{t=0}^{\infty} \beta^t u_c(t) c_t}_{\text{IV}} - \underbrace{\sum_{t=0}^{\infty} \beta^t u_c(t) k_t}_{\text{V}} - \underbrace{\sum_{t=0}^{\infty} \beta^t u_c(t) b_t}_{\text{VI}} \right\} \geq 0$$

There are 6 parts of the equation and we will use CE characterizations to simplify each part. Again, we use the FOCs of consumers and firms in the system of equations (49) - (55) that defines a competitive equilibrium.

First, notice that by (49), we can replace $(1 - \tau_t^h) w_t$ by the MRS between consumption and leisure and rewrite part I as

$$\sum_{t=0}^{\infty} \beta^t u_c(t) (1 - \tau_t^h) w_t h_t = \sum_{t=0}^{\infty} \beta^t u_l(t) h_t$$

By the Euler equation for capital (50), we can rewrite part V as

$$\sum_{t=0}^{\infty} \beta^t u_c(t) k_t = \sum_{t=0}^{\infty} \beta^{t+1} u_c(t+1) R_{t+1}^k k_t$$

By the Euler equation for bonds (51), we can rewrite part VI as

$$\sum_{t=0}^{\infty} \beta^t u_c(t) b_t = \sum_{t=0}^{\infty} \beta^{t+1} u_c(t+1) R_{t+1}^b b_t$$

Now, notice that we can cancel out some terms in part II and part IV, and similarly for part III and part VI:

$$\sum_{t=0}^{\infty} \beta^t u_c(t) R_t^b b_{t-1} - \sum_{t=0}^{\infty} \beta^{t+1} u_c(t+1) R_{t+1}^b b_t = u_c(0) R_0^b b_{-1} \quad (58)$$

$$\sum_{t=0}^{\infty} \beta^t u_c(t) R_t^k k_{t-1} - \sum_{t=0}^{\infty} \beta^{t+1} u_c(t+1) R_{t+1}^k k_t = u_c(0) R_0^k k_{-1} \quad (59)$$

Therefore, we end up with the following equation

$$\mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t (u_c(t)c_t - u_l(t)h_t) \right\} \leq u_c(0)R_0^b b_{-1} + u_c(0)R_0^k k_{-1}$$

which is the PVIC.

Alternative Derivation of PVIC from Government Budget Constraint

Alternatively, we can derive the PVIC from the government budget constraint. Proceed with the same multiplier $\beta^t u_c(t)$ for each period and sum up the government budget constraint, we get:

$$\begin{aligned} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u_c(t) (b_t + \tau_t^h w_t h_t + \tau_t^k (r_t - \delta) k_{t-1} - R_t^b b_{t-1} - g_t) &\geq 0 \\ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u_c(t) \left(- (1 - \tau_t^h) w_t h_t + w_t h_t + \tau_t^k (r_t - \delta) k_{t-1} + b_t - R_t^b b_{t-1} - g_t \right) &\geq 0 \\ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(- u_c(t) (1 - \tau_t^h) w_t h_t \right) + u_c(t) (w_t h_t + \tau_t^k (r_t - \delta) k_{t-1} + b_t - R_t^b b_{t-1} - g_t) &\geq 0 \end{aligned}$$

Substituting (49) into the first part and we get:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ (- u_l(t) h_t) + u_c(t) (w_t h_t + \tau_t^k (r_t - \delta) k_{t-1} + b_t - R_t^b b_{t-1} - g_t) \right\} \geq 0$$

Notice that,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u_c(t) (b_t - R_t^b b_{t-1}) = \mathbb{E}_0 \left(u_c(0)b_0 - u_c(0)R_0^b b_{-1} + \beta u_c(1)b_1 - \beta u_c(1)R_1^b b_0 + \beta u_c(2)b_2 - \beta u_c(2)R_2^b b_1 + \dots \right)$$

And by (51), we have $u_c(t) = \beta u_c(t+1)R_{t+1}^b$, so we can cancel everything except for the term

$$-u_c(0)R_0^b b_{-1}$$

Now, our constraint looks like:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ (- u_l(t) h_t) + u_c(t) (w_t h_t + \tau_t^k (r_t - \delta) k_{t-1} - g_t) \right\} \geq u_c(0)R_0^b b_{-1}$$

Now, by feasibility constraint (54), and F being CRS, we have

$$\begin{aligned} c_t + g_t + k_t &= z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} \\ c_t + k_t - w_t h_t - r_t k_{t-1} - (1 - \delta)k_{t-1} &= -g_t \end{aligned}$$

Now focus on this specific part

$$\begin{aligned} w_t h_t + \tau_t^k (r_t - \delta) k_{t-1} - g_t &= w_t h_t + \tau_t^k (r_t - \delta) k_{t-1} + c_t + k_t - w_t h_t - r_t k_{t-1} - (1 - \delta)k_{t-1} \\ &= \tau_t^k (r_t - \delta) k_{t-1} + c_t + k_t - r_t k_{t-1} - (1 - \delta)k_{t-1} \\ &= c_t + k_t + k_{t-1} (\tau_t^k r_t - \tau_t^k \delta - r_t - 1 + \delta) \\ &= c_t + k_t - (1 + (1 - \tau_t^k)(r_t - \delta)) k_{t-1} \\ &= c_t + k_t - R_t^k k_{t-1} \end{aligned}$$

Then the constraint looks like

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ (-u_l(t) h_t) + u_c(t) (c_t + k_t - R_t^k k_{t-1}) \right\} \geq u_c(0) R_0^b b_{-1}$$

Now, expand

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u_c(t) (k_t - R_t^k k_{t-1}) = \mathbb{E}_0 \left(u_c(0) k_0 - u_c(0) R_0^k k_{-1} + \beta u_c(1) k_1 - \beta u_c(1) R_1^k k_0 + \beta u_c(2) k_2 - \beta u_c(2) R_2^k k_1 + \dots \right)$$

By (50): $u_c(t) = \beta u_c(t+1) R_{t+1}^k$, we can cancel almost everything except

$$-u_c(0) R_0^k k_{-1}$$

Now, our PVIC looks like:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (-u_l(t) h_t + u_c(t) c_t) \geq u_c(0) R_0^k k_{-1} + u_c(0) R_0^b b_{-1}$$

Alongside with the PVIC from consumer's budget constraint, we now get the PVIC with equality

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (-u_l(t) h_t + u_c(t) c_t) = u_c(0) R_0^k k_{-1} + u_c(0) R_0^b b_{-1}$$

2.2.2 Solving Ramsey Planner's Problem

Now, we can solve the Ramsey problem:

$$\max_{c_t, h_t, k_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t, 1 - h_t)$$

subject to

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [u_c(c_t, 1 - h_t)c_t - u_l(c_t, 1 - h_t)h_t] = u_c(0)R_0^k k_{-1} + u_c(0)R_0^b b_{-1} \quad (60)$$

$$z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} - c_t - g_t - k_t = 0 \quad (61)$$

As before, we can set up a Lagrangian

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t, 1 - h_t) + \lambda(u_c(t)c_t - u_l(t)h_t - u_c(0)R_0^k k_{-1} - u_c(0)R_0^b b_{-1}) \right. \\ & \left. + \gamma_t(z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} - c_t - g_t - k_t) \right\} \end{aligned} \quad (62)$$

A Further Investigation into Time Inconsistency

Neither the lecture notes nor the paper are clear about the time-inconsistency issue in the Ramsey planner's problem, but the issue is important. This section examines it further.

By the Bellman optimality principle, (62) can be written recursively as

$$\begin{aligned} V_0(k_{-1}, b_{-1}) = & \max_{c_0, h_0, k_0} u(c_0, 1 - h_0) + \lambda(u_c(0)c_0 - u_l(0)h_0 - u_c(0)R_0^k k_{-1} - u_c(0)R_0^b b_{-1}) \\ & + \gamma_0(z_0 F(k_{-1}, h_0) + (1 - \delta)k_{-1} - c_0 - g_0 - k_0) + \beta \mathbb{E}_0 V_1(k_0, b_0) \end{aligned} \quad (63)$$

From the period-0 perspective, V_1 can be written recursively as

$$\begin{aligned} V_1(k_0, b_0) = & \max_{c_1, h_1, k_1} u(c_1, 1 - h_1) + \lambda(u_c(1)c_1 - u_l(1)h_1 - u_c(0)R_0^k k_{-1} - u_c(0)R_0^b b_{-1}) \\ & + \gamma_1(z_1 F(k_0, h_1) + (1 - \delta)k_0 - c_1 - g_1 - k_1) + \beta \mathbb{E}_1 V_2(k_1, b_1) \end{aligned} \quad (64)$$

Equations (63) and (64) represent fundamentally different problems. For instance, the envelope condition for V_0 with respect to k is

$$\frac{\partial V_0}{\partial k_{-1}} = -\lambda u_c(0)R_0^k + \gamma_0(z_0 F_k(k_{-1}, h_0) + 1 - \delta) \quad (65)$$

whereas the envelope condition for V_1 with respect to k is

$$\frac{\partial V_1}{\partial k_0} = \gamma_1(z_1 F_k(k_0, h_1) + 1 - \delta) \quad (66)$$

The difference is due to the presence of k_{-1} and b_{-1} in the period-0 problem. As long as $k_{-1}, b_{-1} > 0$, the first-order and envelope conditions for the period-0 problem (63) are affected by the $u_c(0)R_0^k k_{-1}$ and $u_c(0)R_0^b b_{-1}$ terms. Moreover, from the period-0 perspective, the Ramsey planner anticipates that these terms in the implementability constraint should no longer matter when determining the optimal allocation from period 1 onward.

However, this is not true once the Ramsey planner reaches period 1. At that point, the planner takes k_0 and b_0 , both of which were predetermined in period 0, as given. Following the same steps as above yields a PVIC from the period-1 perspective:

$$\mathbb{E}_1 \sum_{t=1}^{\infty} \beta^t (-u_l(t)h_t + u_c(t)c_t) = u_c(1)R_1^k k_0 + u_c(1)R_1^b b_0 \quad (67)$$

This leads the period-1 planner to solve a problem analogous to (63) rather than (64). Thus, the planner will change the period-0 plan for period-1 allocations upon reaching period 1; without commitment, the period-0 plan is time-inconsistent.

Intuitively, why does time inconsistency arise? The period-0 Ramsey planner enters with initial capital k_{-1} and bonds b_{-1} and can choose an allocation c_0 that changes the marginal utility of consumption in period 0, and hence the “value” of those initial assets. However, from the period-0 perspective, using the PVIC (57) amounts to assuming that the planner will not or cannot alter the value of capital and bonds by changing the marginal utility of consumption in periods $t \geq 1$.

To resolve the time inconsistency, we focus on the optimal Ramsey plan *under commitment*; that is, we assume that the Ramsey planner commits to the period-0 plan. Furthermore, we disregard the planner’s ability to account for the effect of $u_c(0)$ on the right-hand side of the PVIC, so the planner treats that side of the constraint as a constant:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [u_c(c_t, 1 - h_t)c_t - u_l(c_t, 1 - h_t)h_t] = A_0$$

that is, the Ramsey planner no longer attempts to manipulate the value of initial capital and bonds by changing $u_c(0)$.

Solving Ramsey Planner’s Problem under Commitment

Now, we derive the first-order conditions:

$$u_c(t) + \lambda(u_{cc}(t)c_t + u_c(t) - u_{lc}(t)h_t) = \gamma_t \quad (68)$$

$$-u_l(t) + \lambda(-u_{cl}(t)c_t + u_{ll}(t)h_t - u_l(t)) + \gamma_t(z_t F_h(t)) = 0 \quad (69)$$

$$\beta \mathbb{E}_t \left(\gamma_{t+1} (z_{t+1} F_k(t+1) + 1 - \delta) \right) = \gamma_t \quad (70)$$

The transversality condition is

$$\lim_{t \rightarrow \infty} \beta^t u_c(t) k_t = 0$$

Proposition 6. *The solution to the Ramsey problem is sequences $\{c_t, h_t, k_t, \gamma_t\}_{t=0}^{\infty}$ and a constant λ that satisfies (68), (69), (70), the PVIC (60), the resource constraint (61), and a transversality condition $\lim_{t \rightarrow \infty} \beta^t u_c(t) k_t = 0$, given k_{-1} .*

Back Out Prices and Policies from Allocations

Once we obtain the optimal Ramsey allocation, prices can be calculated from the firm's FOCs:

$$r_t = z_t F_k(k_{t-1}, h_t)$$

$$w_t = z_t F_h(k_{t-1}, h_t)$$

Labor income taxes follow from the intratemporal Euler equation:

$$1 - \tau_t^h = \frac{u_l(c_t, 1 - h_t)}{w_t u_c(c_t, 1 - h_t)} = \frac{u_l(t)}{u_c(t) \cdot z_t F_h(t)}$$

Government bonds are defined through a process similar to the derivation of the PVIC:

$$b_t = \mathbb{E}_t \left\{ \sum_{s=t+1}^{\infty} \beta^{s-t} \frac{u_c(s)c_s - u_l(s)h_s}{u_c(t)} \right\} - k_t$$

Lastly, we are left with

$$u_c(t) = \beta \mathbb{E}_t [u_c(t+1) R_{t+1}^k]$$

$$u_c(t) = \beta \mathbb{E}_t [u_c(t+1) R_{t+1}^b]$$

to define R_t^k , i.e. to find τ_t^k , and R_t^b . There is an indeterminacy between state-contingent capital income taxes and state-contingent returns on government bonds. There are 2 alternatives to fix this:

- assume capital income tax is not state-contingent, i.e. τ_k is pre-determined at time $t - 1$
- assume the return on bond is not state-contingent, i.e. R_t^b is pre-determined at time $t - 1$

2.3 Social Planner's Problem

Social planner's problem here is

$$\max_{c_t, h_t, k_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t, 1 - h_t)$$

subject to

$$z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} - c_t - g_t - k_t = 0$$

Lagrangian looks like:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t, 1 - h_t) + \zeta_t (z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} - c_t - g_t - k_t) \right\}$$

FOC yields:

$$\begin{aligned} u_c(t) &= \zeta_t \\ u_l(t) &= \zeta_t z_t F_h(t) \\ \zeta_t &= \beta \zeta_{t+1} \mathbb{E}_t (z_{t+1} F_k(k_t, h_{t+1}) + 1 - \delta) \\ z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} &= c_t + g_t + k_t \end{aligned}$$

Then the solution can be characterized by:

$$\begin{aligned} \frac{u_c(t)}{u_l(t)} &= \frac{1}{z_t F_k(t)} \\ u_c(t) &= \beta \mathbb{E}(u_c(t+1) z_{t+1} F_k(t+1) + 1 - \delta) \\ z_t F(k_{t-1}, h_t) + (1 - \delta)k_{t-1} &= c_t + g_t + k_t \end{aligned}$$

And transversality condition

$$\lim_{t \rightarrow \infty} \beta^t u_c(t) k_t = 0$$

2.4 Completeness of Tax System

From the social planner's problem, we see that there are 2 margins in this economy:

- intratemporal margin: consumption today vs leisure today
- intertemporal margin: consumption today vs consumption tomorrow

To implement the Ramsey allocations, we have 2 instruments: labor income tax and capital income tax. This is a complete system, as there are 2 margins, and 2 instruments that can create wedges between those margins.

To illustrate this, recall the Ramsey problem characterizations:

$$\begin{aligned} u_c(t) + \lambda(u_{cc}(t)c_t + u_c(t) - u_{lc}(t)h_t) &= \gamma_t \\ u_l(t) - \lambda(-u_{cl}(t)c_t + u_{ll}(t)h_t - u_l(t)) &= \gamma_t(z_t F_h(t)) \end{aligned}$$

Combining both we get:

$$\begin{aligned} \frac{u_c(t) + \lambda(u_{cc}(t)c_t + u_c(t) - u_{lc}(t)h_t)}{u_l(t) - \lambda(-u_{cl}(t)c_t + u_{ll}(t)h_t - u_l(t))} &= \frac{1}{z_t F_h(t)} \\ \frac{(1 + \lambda)u_c(t) + \lambda(u_{cc}(t)c_t - u_{lc}(t)h_t)}{(1 + \lambda)u_l(t) + \lambda(u_{cl}(t)c_t - u_{ll}(t)h_t)} &= \frac{1}{z_t F_h(t)} \end{aligned}$$

It is possible to perform what we had in previous section, to see that by algebraic manipulations, we can create a wedge between $\frac{u_c(t)}{u_l(t)}$ and $\frac{1}{z_t F_h(t)}$, i.e. a wedge between $MRS_{c,l}$ and $MRT_{c,l}$.

Another important result is that it is possible to **replace the labor income tax with a consumption tax**, and this will not change the Ramsey allocations.

The new budget constraint for consumers becomes:

$$(1 + \tau_t^c)c_t + k_t + b_t = w_t h_t + R_t^k k_{t-1} + R_t b_{t-1}$$

The new intratemporal Euler equation becomes;

$$\frac{u_l(t)}{u_c(t)} = \frac{w_t}{1 + \tau_t^c}$$

Recall the labor income tax result, we see that if we set τ_t^c so that:

$$1 - \tau_t^h = \frac{1}{1 + \tau_t^c}$$

The Ramsey problem allocations are exactly the same.

Similarly, it is possible to have all three taxes, introducing one more indeterminacy into the problem.

2.5 Capital Taxation Credit

One of the most important results from this setup is the zero-capital taxation result. At the steady state, the FOC on k_t , (70), implies

$$\beta(F_k + 1 - \delta) = 1$$

And in a steady-state competitive equilibrium, Euler equation implies

$$u_c = \beta u_c [1 + (1 - \tau^k)(F_k - \delta)] \implies 1 = \beta [1 + (1 - \tau^k)(F_k - \delta)]$$

Combining both, we get the Ramsey optimal plan requires the capital tax to be 0, namely $\tau^k = 0$.

Note that from the wedge interpretation from the intertemporal margin, we have the *MRT* of consumption today vs consumption tomorrow is $F_k + 1 - \delta$, whereas the *MRS* of such being $\frac{1}{\beta}$.

In the long run, Ramsey planner wants to keep $MRT = MRS$. The no long-run intertemporal distortions is a more general result than the zero-capital taxation result.